Chapter 11: Consumer Choice Theory: Utility and Indifference Curves

Principles of Microeconomics Curriculum

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Abstract

This chapter provides a rigorous, pedagogical treatment of Chapter 11: Consumer Choice Theory: Utility and Indifference Curves, benchmarked to standard principles curriculum.

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1 Learning Outcomes and Key Terms

Students should represent a budget constraint, identify an affordable choice, and explain how prices and income shape a consumer's decision. These objectives support C2001-1 and C2001-3. Key terms include budget set, utility, indifference curve, marginal utility, marginal rate of substitution, substitution effect, and income effect.

2 A Monthly Budget and Two Goods

Suppose a student allocates a fixed monthly budget between meals (X) and transit trips (Y). The budget is limited, so buying more of one good means giving up some of the other. The consumer's choice depends on prices, income, and preferences; the model does not claim that every person calculates derivatives before buying lunch. It organizes the trade-offs implied by those constraints.

3 The Budget Line: Geometric Description

Place meals X on the horizontal axis and transit trips Y on the vertical axis. The budget line slopes downward. Its horizontal intercept is the number of meals affordable if all income is spent on meals; its vertical intercept is the number of transit trips affordable if all income is spent on transit. A rise in income shifts the budget line outward in parallel when prices are fixed. A change in one price pivots the line around the intercept for the other good.

The budget constraint is

$$P_x X + P_y Y = I, \qquad Y = \frac{I}{P_y} - \frac{P_x}{P_y} X. \tag{1}$$

The slope is $-P_x/P_y$: it is the number of transit trips that must be forgone to purchase one more meal while staying on the same budget line.

4 Preferences, Utility, and Indifference Curves

An indifference curve collects bundles that provide the same ordinal utility. A higher curve represents a preferred bundle when preferences are monotonic. With well-behaved preferences, indifference curves slope downward and are convex to the origin: as a person has more of X relative to Y, they are generally willing to give up fewer units of Y for another unit of X.

Marginal utility (MU_x) is the change in utility from a small increase in X, holding Y fixed. The marginal rate of substitution is the absolute slope of an indifference curve:

$$MRS_{XY} = \frac{MU_x}{MU_y}. \tag{2}$$

Utility numbers are a representation for ranking choices, not a literal measurement of happiness that can be compared across people.

5 Choice at a Tangency: Worked Example

Let income be $I = 60$, with $P_x = 4$ and $P_y = 3$. The consumer has Cobb-Douglas utility $U(X, Y) = XY$. Since $MU_x = Y$ and $MU_y = X$, the tangency condition $MRS = P_x/P_y$ implies $Y/X = 4/3$. Combining this with the budget constraint gives $X^* = 7.5$ meals and $Y^* = 10$ transit trips. The bundle exhausts the budget: $4(7.5) + 3(10) = 60$.

For an interior optimum with smooth, convex preferences:

$$\frac{MU_x}{MU_y} = \frac{P_x}{P_y} \quad \Longleftrightarrow \quad \frac{MU_x}{P_x} = \frac{MU_y}{P_y}. \tag{3}$$

At a corner solution, the consumer chooses only one good; the tangency equality need not hold. Perfect substitutes, perfect complements, and nonstandard preferences can also change the shape of the solution.

6 Price Changes: Substitution and Income Effects

If the price of X falls, the budget line pivots outward along the X axis. The total change in the chosen amount of X can be separated conceptually into a substitution effect and an income effect. The substitution effect reflects the change in relative prices while holding purchasing power conceptually constant; the income effect reflects the change in real purchasing power. For a normal good, both effects of a price decrease usually raise quantity demanded. For an inferior good, the income effect works in the opposite direction and may partly offset the substitution effect. A parallel budget shift is caused by a change in income at fixed prices; it is not the price-change income effect itself.

Clear It Up: The Price Change Pivots the Budget Line

A change in P_x changes the budget line's slope and rotates it around the Y intercept. A change in income shifts the entire budget line in parallel when both prices remain fixed. Keep these two geometric changes distinct.

7 FRED Data and Real Purchasing Power

FRED series CPIAUCSL is the Consumer Price Index for All Urban Consumers: All Items in U.S. City Average. It is monthly, seasonally adjusted, and indexed to 1982-1984=100; its September 2026 observation is 336.298. The CPI is a broad price index, not the exact cost of every household's basket. It can help convert a nominal dollar amount to a broad real-dollar comparison, but it does not identify an individual's preferences or their personal inflation rate.

8 Check Your Understanding

1. With $I = 60$, $P_x = 4$, and $P_y = 3$, what are the two intercepts of the budget line?
2. Why does the budget line pivot rather than shift in parallel when only P_x changes?
3. Is utility measured in directly comparable units across people? Explain.

Answers: (1) $X = 15$ and $Y = 20$. (2) The X intercept changes while the Y intercept remains I/P_y. (3) No; ordinal utility represents an individual's ranking of bundles.